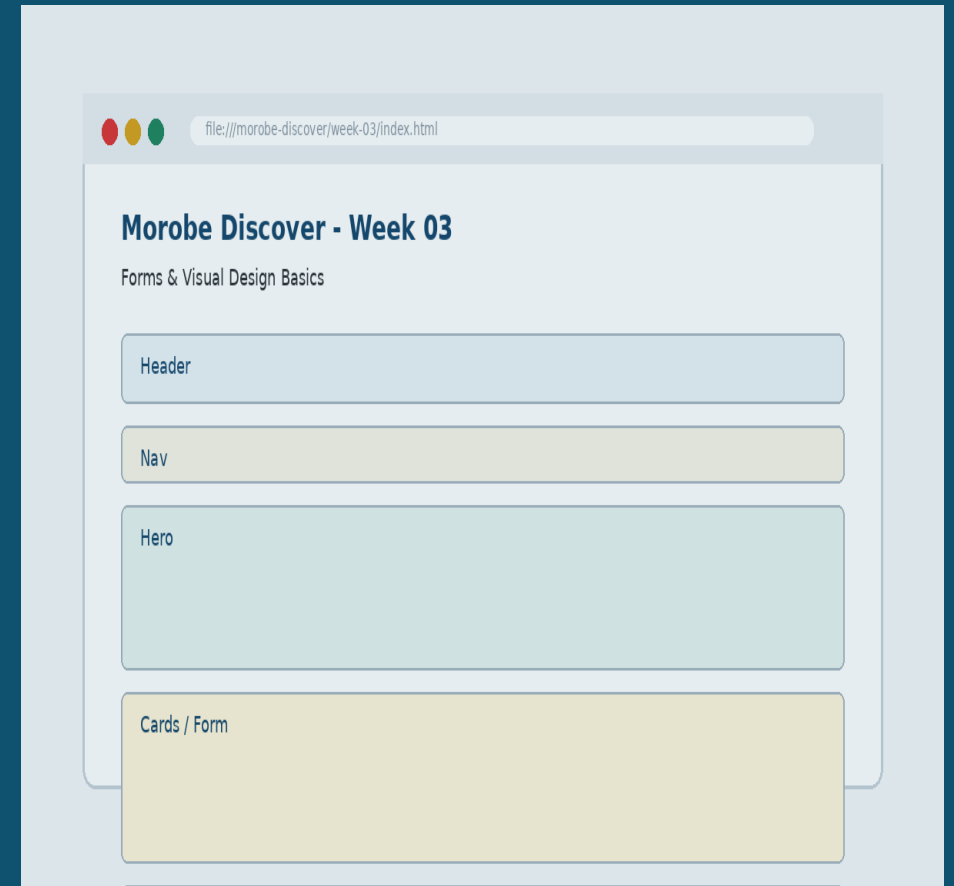


Week 03

Forms & Visual Design Basics

Implement user data collection and apply visual design principles to control user attention.



Learning outcomes

Implement user data collection and apply visual design principles to control user attention.

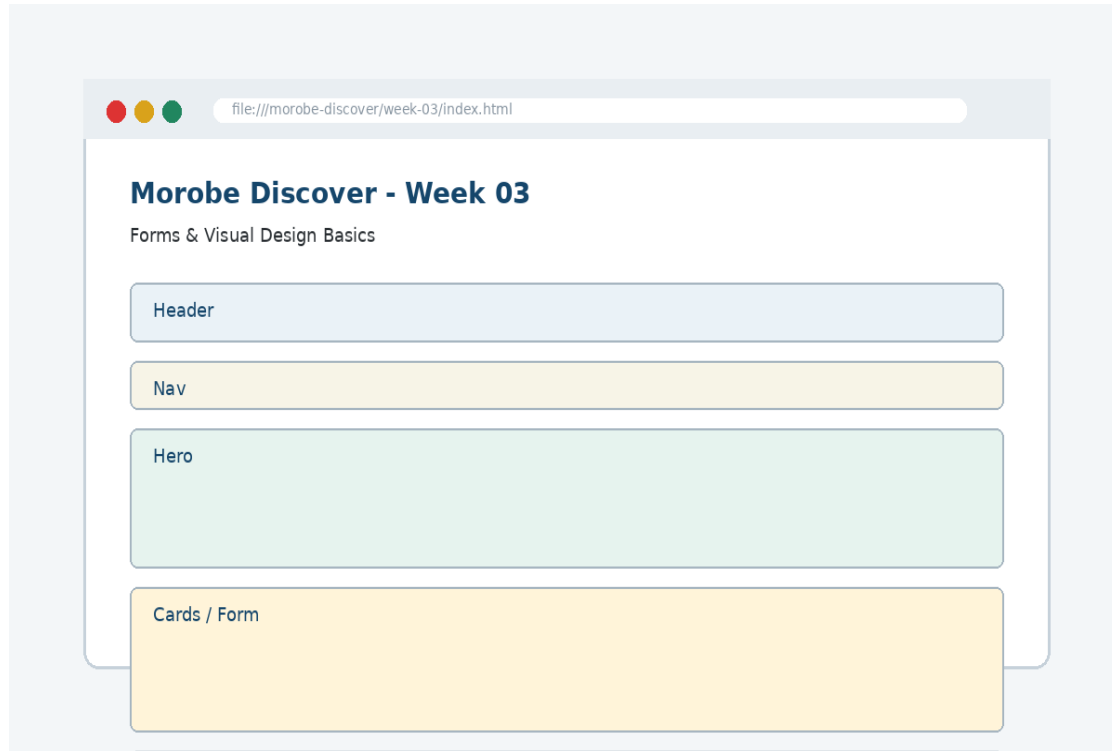
Explain the core programming/design concept for this week

Read and modify key code patterns

Test the weekly project increment

Prepare evidence for GitHub submission

Today's problem



Why it matters

Forms are conversations with users; visual design should make that conversation easy to follow.

Project before -> after

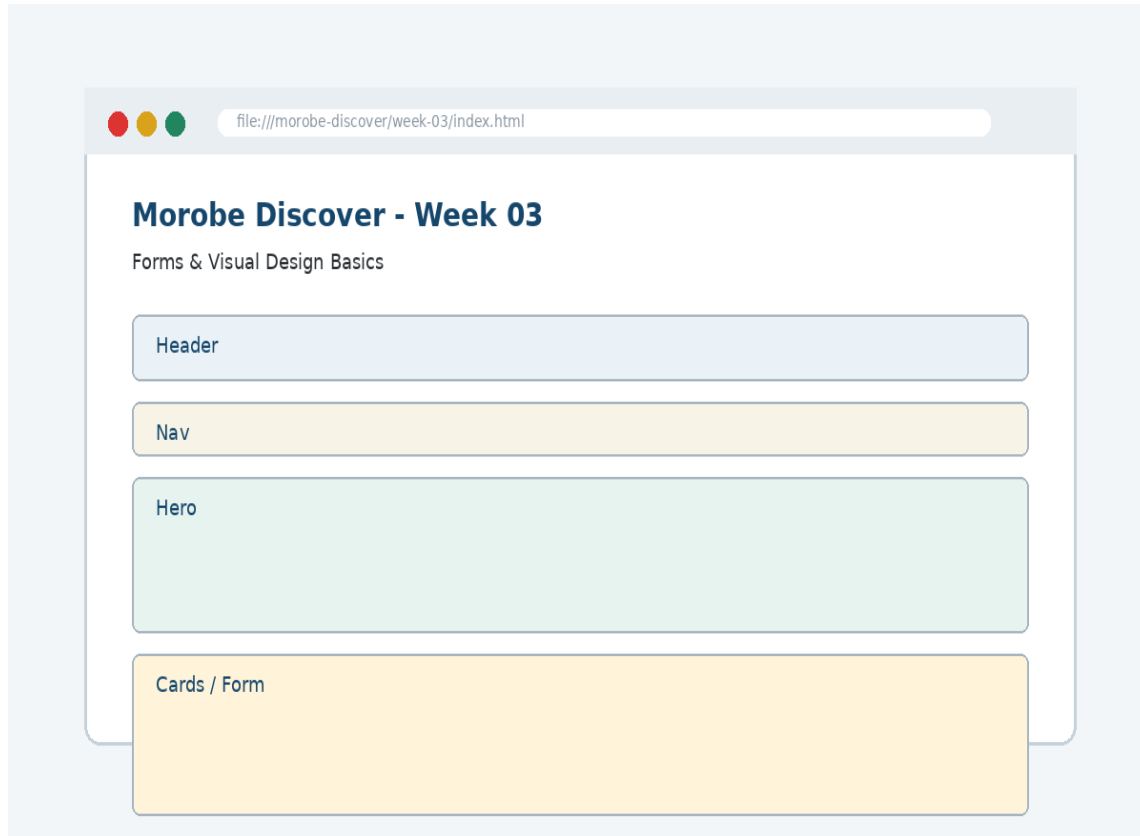
Before Week 03

Week 02 created a semantic homepage and page regions.

After Week 03

The site includes a basic planning form and simple visual hierarchy through CSS.

What we build this week



Create the Plan a Trip form
Use labels and suitable input types
Add required fields and helpful placeholder text
Create basic CSS variables
Apply typography, colour and spacing

Core principle

Forms are conversations with users; visual design should make that conversation easy to follow.

Keep in mind

A good web developer can point to the code, the browser result, and the user reason behind the decision.

Key concepts

Concept 1

Labels connect meaning to inputs

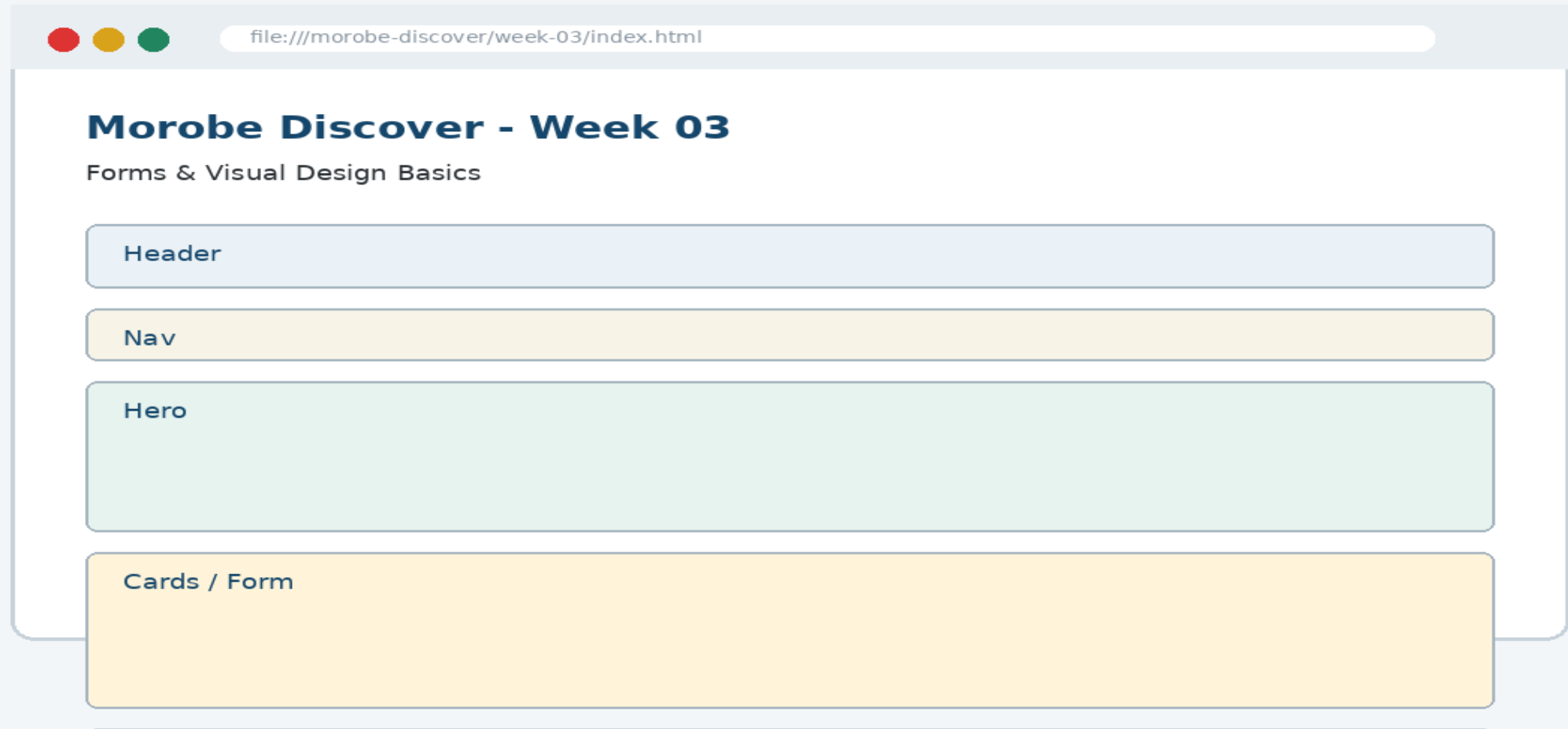
Concept 2

Input types improve mobile keyboards and validation

Concept 3

Typography, colour and spacing create hierarchy

Project anatomy



Key code pattern A

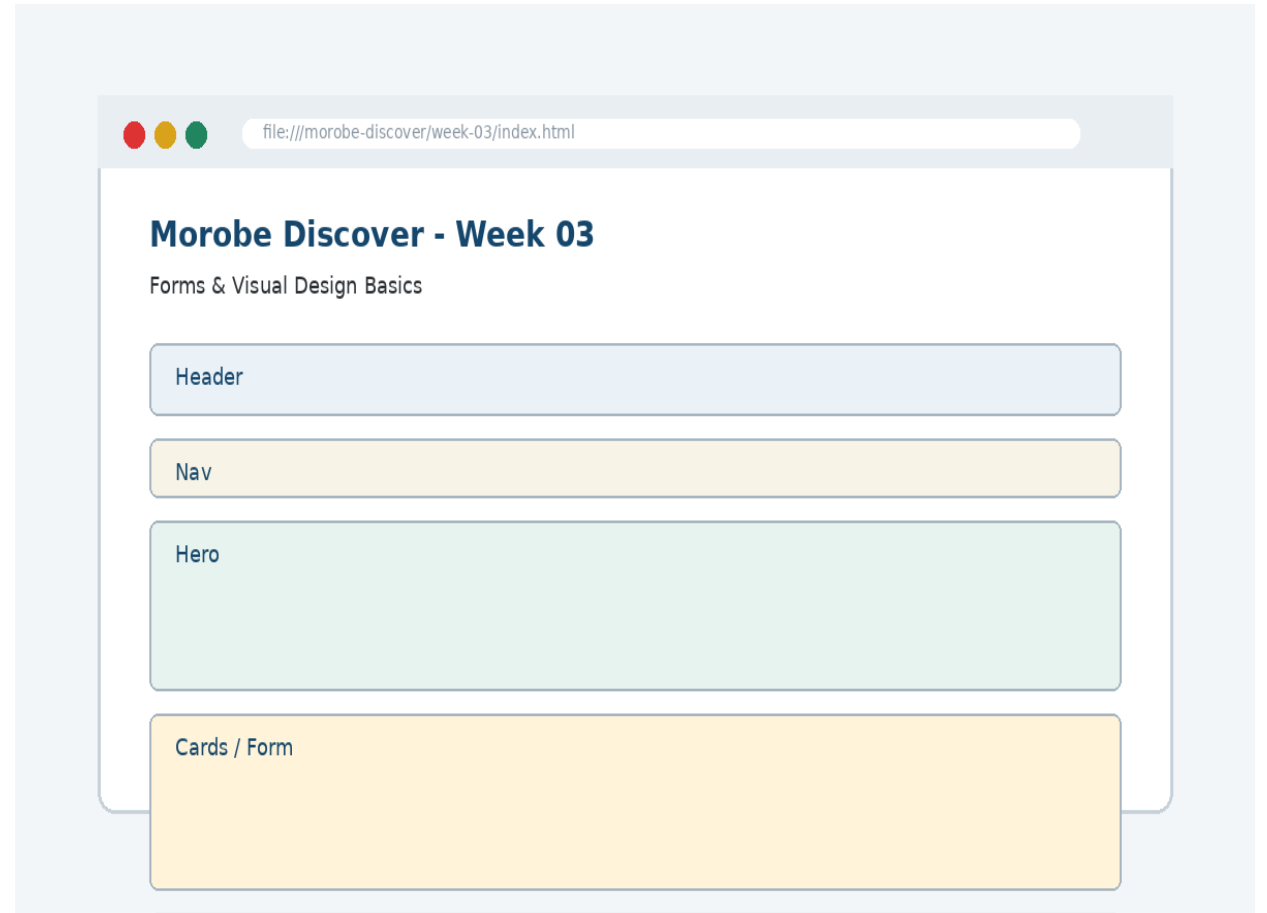
```
<label for="visitorName">Your name</label>  
<input id="visitorName" name="visitorName"  
type="text" required>
```

Explain it

Which file contains this code?
Which line creates the visible change?
What should appear in the browser?

Code -> visual result

```
<label for="visitorName">Your  
name</label>  
<input id="visitorName"  
name="visitorName" type="text" required>
```



What changed?

Project file changed for Forms & Visual Design Basics
The browser output is now closer to the required weekly evidence
The change supports a user task or quality requirement
The change can be tested and committed

Key code pattern B

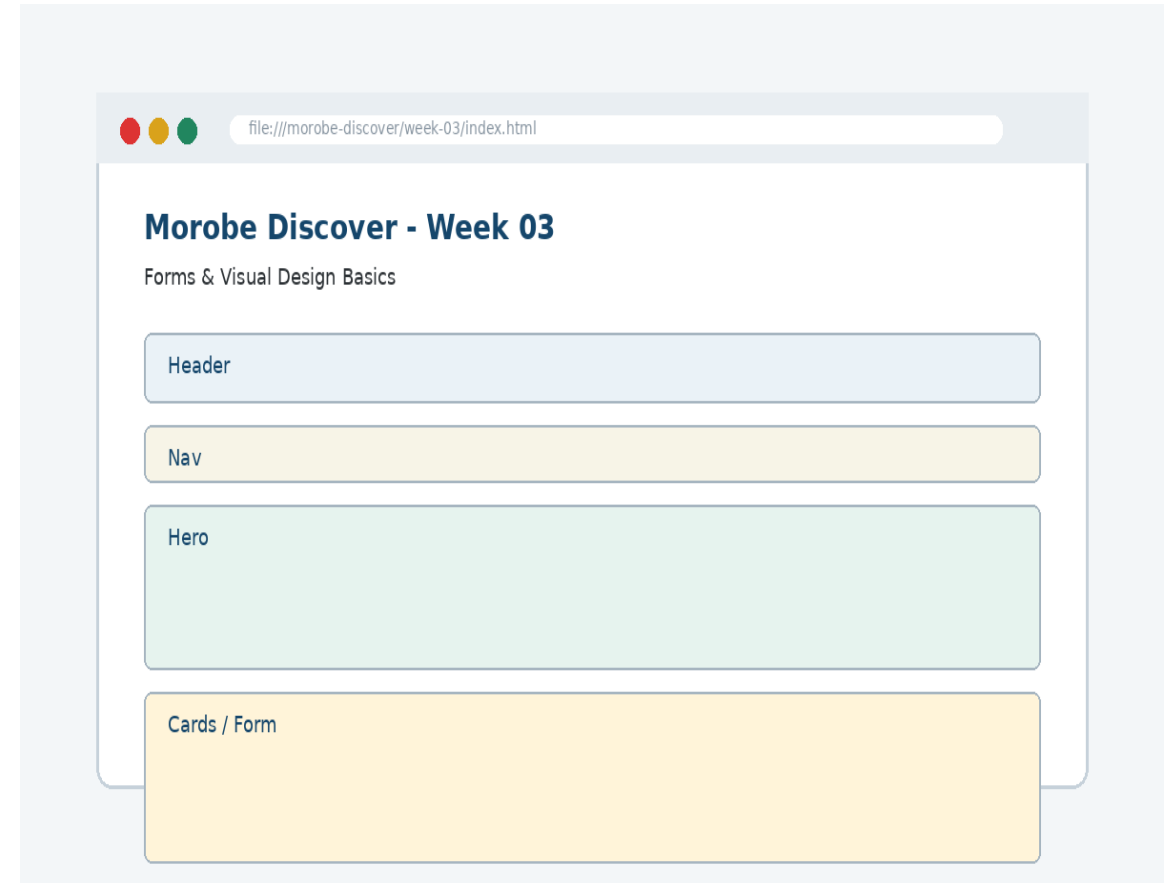
```
form {  
  max-width: 720px;  
  display: grid;  
  gap: 1rem;  
}
```

Why this pattern is needed

The second pattern usually applies the same idea to a different component, state, layout or data flow.

Apply it to Morobe Discover

Create the Plan a Trip form
Use labels and suitable input types
Add required fields and helpful placeholder text
Create basic CSS variables



Compare the approaches

Weak approach

Memorise the definition
Copy code without understanding
No browser evidence
No test note

Professional approach

Explain the concept
Apply the correct code pattern
Inspect the result
Commit and describe the evidence

Common mistake

Mistake

The page appears unchanged, but the student assumes the feature is complete.

Correction

Check the correct file, refresh the browser, inspect DevTools, and verify the expected output.

Why does this fail?

```
// Broken checkpoint
// File changed, but the browser result does not
change.
// Possible causes:
// - wrong file
// - file not linked
// - syntax error
// - browser not refreshed
```

Class question

Which cause would you check first, and why?

Fix the problem

- Confirm the browser is opening the correct file
- Check links to CSS/JS/data files
- Inspect the console for errors
- Use a small working example
- Test again before committing

Class activity

Activity - 7 minutes

Identify which form fields are essential and which would create unnecessary friction.

Work in pairs

Point to the project evidence

Prepare one answer to share

Implementation steps 1-3

Create the Plan a Trip form

Use labels and suitable input types

Add required fields and helpful placeholder text

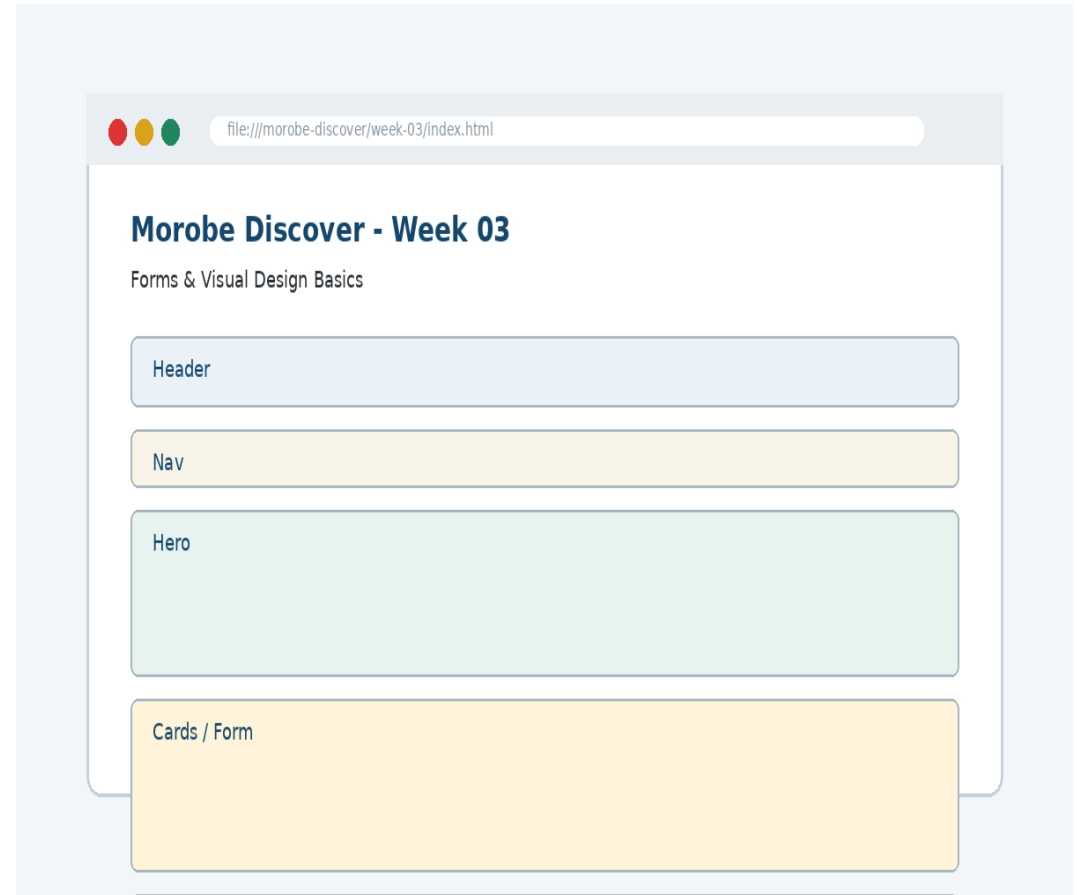
Implementation steps 4-5

Create basic CSS variables
Apply typography, colour and spacing

LIVE DEMO

Demo checklist

1. Open the current project
2. Locate the affected file
3. Add or modify the key code
4. Refresh the browser
5. Inspect the result
6. Explain the change



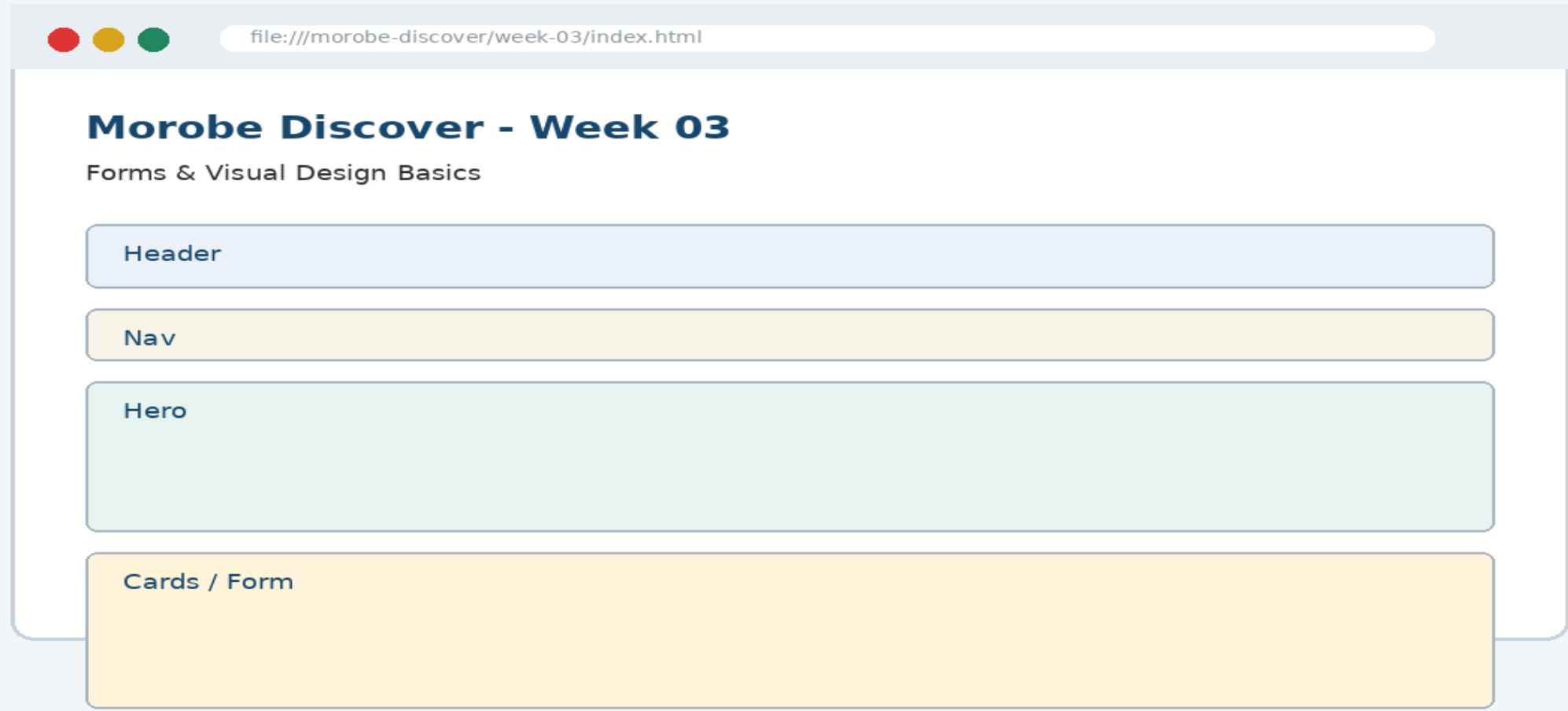
Predict the output

```
form {  
  max-width: 720px;  
  display: grid;  
  gap: 1rem;  
}
```

Prediction prompt

What should change on screen when this code is applied correctly?

Browser result



Testing the feature

- Try submitting an empty form
- Check every input has a visible label
- Verify focus moves logically
- Check text contrast and spacing

Inspect with browser tools

Inspection path

Right click -> Inspect
Elements panel -> structure/classes
Console -> JavaScript errors
Network/Application -> data/storage when relevant

Evidence to save

Screenshot
Short test note
Changed files
Git commit message

Git evidence

```
git add .  
git commit -m "Week 03 Forms & Visual Design Basics  
project increment"
```

Why it matters

Git history shows progression from Week 01 to final deployment. It is part of the assessment evidence.

Complete source code

Repository:
`https://github.com/YOUR-USERNAME/is229-morobe-discover`

Weekly folder:
`/week-03/`

Complete project:
`/complete-project/`

Reminder

Do not paste a whole file into a slide. Use the repository for complete code.

Mini challenge

Challenge

Modify one part of the Week 03 implementation and explain what changed in the browser.

State the file edited

State the line/section changed

Show the before and after result

What counts as evidence?

Technical evidence

Changed code files
Browser screenshot
Test note
Git commit

Understanding evidence

Code explanation
Why this method fits
Connection to previous week
Connection to next week

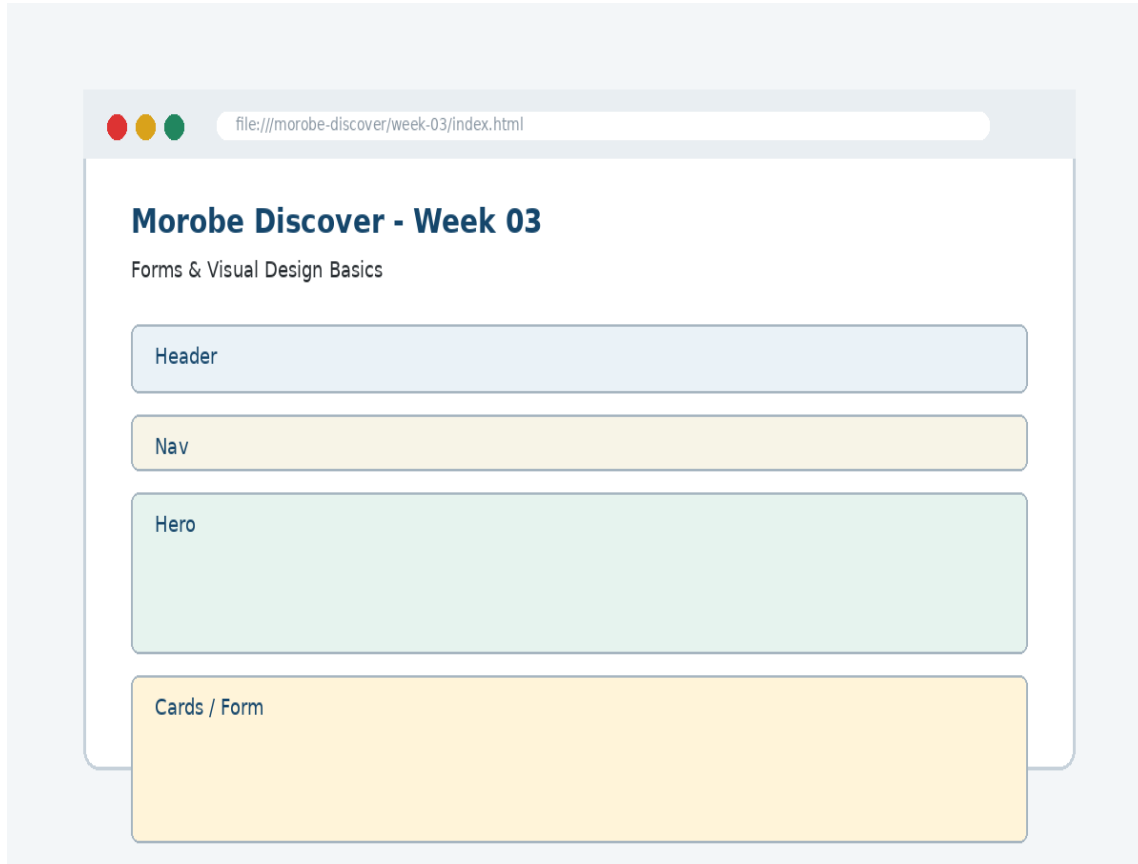
Project lab handoff

Create the Plan a Trip form
Use labels and suitable input types
Add required fields and helpful placeholder text
Create basic CSS variables
Apply typography, colour and spacing

Definition of done

Week 03 feature is implemented
Project still preserves earlier week features
Browser result matches the expected output
Test note and screenshot are saved
Git commit message is meaningful

What we built today



The site includes a basic planning form and simple visual hierarchy through CSS. Evidence: Form fields, input types, labels, visual hierarchy, typography and colour rules

Technology: HTML / CSS

Lecture summary

Forms are conversations with users; visual design should make that conversation easy to follow.

Labels connect meaning to inputs

Input types improve mobile keyboards and validation

Testing and Git are part of development, not extras

Prepare for next class

Finish this week's lab before the next class
Read the next topic in advance
Bring the current project ready to extend
Be ready to explain one code decision

Exit question

How did Week 03 improve Morobe Discover, and what will the next week build on?